Urban service allocation: consolidated traffic and sensitivity specification

Selective-discreteness urban review

27 October 2026

Purpose and status. This document applies the closed traffic-policy discussion, including feedback 10. It defines a discrete continuous-mass model for methodological review; it does not report an implemented or validated simulator. Unadmitted mass has no binding future-lane target. The earlier waiting-quota construction and its derivatives are removed from the current model and preserved in the dated archive. Numerical settings and source-native rival implementation qualification remain later work.

1 Traffic world, action and physical ownership

Retain one isolated four-leg junction, fixed routes and four protected effective-service groups: NS through/right, NS left, EW through/right, EW left. With physical capacity C_ℓ and saturation $s_{\ell m}$,

$$g_p \geq 0, \quad \sum_{p=1}^{4} g_p = 1, \qquad \mu_{\ell m}(g) = \min\{C_\ell, s_{\ell m} g_{p(m)}\}. \tag{1}$$

Mass is continuous vehicle-equivalent quantity; C, s, μ are veh-eq/s. A feasible direction preserves the simplex. Route labels, permissions, topology and source-order skeleton are structural. No phase waveform, integerization, microscopic lane changing or endogenous route choice is introduced. Control uses constant g during the declared control window, then a common candidate-independent clearance allocation.

For the selected bay scenario, represent the approach as

$$U \longrightarrow F \longrightarrow \begin{cases} B & \text{left-only bay,} \\ T & \text{shared left/through/right continuation.} \end{cases} \tag{2}$$

U is the aggregate upstream road; F is an actual common lane inside the local zone, before the fork. B, T begin at that fork. Their lengths are equal in the symmetric core, and approach path length is $L_U + L_F + L_B = L_U + L_F + L_T$. Every section owns its storage; these are disjoint road pieces, not overlapping copies of one link.

V0 uses movement-exclusive full-length paths. V1 uses a shared lane. V2/V3 share the chosen bay geometry and permissions: V2 suppresses order-sensitive obstruction by construction, whereas V3 tests it. V4 supplies predeclared held-out variation. Permissions may differ between scenario families, but not between compared models or paired order cases of the same scenario.

Two meanings of order. An unassigned link-end or external-source pool has an ordered processing record. A realized local lane has a physical FIFO sequence. The former does not establish a single physical lane position for every packet. The latter does establish front blocking. Classification follows the represented road; an occupied F cannot be relabeled a pool to permit passing. The graph in (2) has no direct U–B or U–T shortcut.

2 Propagation and the step clock

Each physical section retains cumulative counts $N^+_{e,k}, N^-_{e,k}$ and occupancy $n_{e,k} = N^+_{e,k} - N^-_{e,k}$ at $t_k = k\Delta t$. Use triangular-FD capacity C_e, free-flow delay τ^f_e, backward delay τ^b_e and storage $K_e = C_e(\tau^f_e + \tau^b_e)$, with $\Delta t \leq \min_e(\tau^f_e, \tau^b_e)$. Road capacities, delays and legal saturation rates are finite and positive. An

incompatible short section requires a compatible discretization; no fictitious delay, clipping or smoothing is inserted.

With empty prehistory and $\theta = (t - t_h)/\Delta t$,

$$\mathcal{H}_e^{\pm}(t) = (1-\theta)N_{e,h}^{\pm} + \theta N_{e,h+1}^{\pm}, \qquad t_h \le t \le t_{h+1}. \tag{3}$$

The inherited pre-state one-step limits are

$$A_{e,k} = [\mathcal{H}_e^{+}(t_{k+1} - \tau_e^f) - N_{e,k}^{-}]_+, \qquad S_{e,k} = \min(C_e\Delta t, A_{e,k}), \tag{4}$$

$$R_{e,k} = \min\{C_e\Delta t,\, [K_e + \mathcal{H}_e^{-}(t_{k+1} - \tau_e^b) - N_{e,k}^{+}]_+\}. \tag{5}$$

A is ready resident mass for this update; S, R are mass allowances, not instantaneous rates. Queries use already stored boundaries. New inflow cannot traverse its receiving section in the same update.

This is a timestep model. The internal head-service calculation produces endpoint totals; it does not add propagation-history knots. Readiness uses the inherited discrete sending convention, rather than claiming every selected resident was ready at every instant in the step. No event-history refinement is adopted.

These equations and the left-interval objective are inherited from the approved repository M0 transcription, checked against the local LTM and objective code. Their application to the internal section graph is the declared urban composition, not existing implemented M2 behavior.

3 Ledger, eligibility and lane membership

Store positive-mass cohorts, removing exact zero entries without a cutoff. A cohort stores route-table reference, route position, continuous mass, current owner and affine eligibility provenance. Current movement is projected from its route. Internal $U/F/B/T$ crossings preserve route position; the actual junction crossing advances it. Future movement sequences are not duplicated in grid entries.

Only actual admission establishes receiving-lane membership. Unadmitted mass stays with its old owner without a binding future-lane vector. Traffic already in F occupies that common lane, but has no frozen B/T target. Traffic already in B or T cannot be reassigned to improve balance.

Retain the exact source order of the unserved record. Before evaluating an operator, coalesce exactly compatible adjacent representation fragments: same route/position, owner, permissions and contiguous affine eligibility with the M1 exact line condition. No tolerance-based mixing or route merging is used. Serialization changes must preserve predictions; they do not define a new physical traffic unit.

The movement grid is a view of the ordered ledger:

$$M_{cm} = \sum_{r:m(r)=m} |I_r \cap B_c|, \tag{6}$$

where I_r, B_c are intervals in cumulative resident mass. Rank is not metres; mixed composition does not reconstruct exact head order. Cells cover the full resident mass-rank range without truncating the ledger. Storage, travel and spillback come from physical histories.

Ready candidates are not a universal sending prefix. Form candidates from ready mass A, cutting an eligibility segment exactly if needed. For an unassigned pool, scan these candidates and spend at most S in total. Do not first keep only the earliest S mass: an earlier blocked movement must not hide later ready mass that has independent access. For a physical lane, the access rule below forces actual served mass to be a prefix.

This is an explicit adaptation of M1's prefix extraction. After a non-prefix pool selection, retained cohorts keep original eligibility provenance. Every past departure was eligible when served, and eligibility is monotone. Total eligible residual mass therefore remains $\mathcal{H}^{+}(t_{k+1} - \tau^f) - N_k^{-}$ even if eligible departures were not a global processing-order prefix. The adaptation is not claimed to exist in current M1 code, nor to establish passing on a represented single lane.

4 Allocation, unassigned residuals and physical access

Let a be the sending owner, r a ready cohort of mass q_r and movement m_r, and ℓ a receiving lane/section. Permission $P_{m\ell} \in \{0,1\}$ and reachability $E_{r\ell} \in \{0,1\}$ encode actual paths/access, not a future green plan.

Initialize residual sending $s = S_{a,k}$, residual receiving $R_\ell^{\mathrm{rem}} = R_{\ell,k}$ and perceived loads $h_\ell = n_{\ell,k}$. Process cohorts in recorded order. Let $\mathcal{B}$ contain movements already left partly unserved in this scan. Each sending owner has its own scan and blocked-movement set; one inbound leg does not block another. If $m_r \in \mathcal{B}$ set $V_r = x_{r\ell} = 0$; otherwise set

$$
\begin{aligned}
u_{r\ell} &= P_{m_r\ell} E_{r\ell} R_\ell^{\mathrm{rem}}, \qquad V_r = \min\{q_r, s, \sum_\ell u_{r\ell}\}, \\
x_r &= \operatorname*{arg\,min}_{0 \le x_{r\ell} \le u_{r\ell},\ \sum_\ell x_{r\ell} = V_r} \frac{1}{2} \sum_{\ell : P_{m_r\ell} = 1} (h_\ell + x_{r\ell})^2, \\
x_{r\ell} &= \min\{u_{r\ell}, [\lambda_r - h_\ell]_+\}, \qquad \sum_\ell x_{r\ell} = V_r.
\end{aligned} \tag{7}
$$

Illegal lanes have zero addition. Strict convexity selects unique additions; the threshold itself need not be unique on a capped plateau. For $V = 0$, use $x = 0$. Commit only actual entry:

$$
s \leftarrow s - V_r, \quad R_\ell^{\mathrm{rem}} \leftarrow R_\ell^{\mathrm{rem}} - x_{r\ell}, \quad h_\ell \leftarrow h_\ell + x_{r\ell}, \quad q_r^{\mathrm{rem}} = q_r - V_r. \tag{8}
$$

If $q_r^{\mathrm{rem}} > 0$, retain it at its original position and add m_r to $\mathcal{B}$ for this update. Later cohorts of that movement wait; other movements are considered. Reset $\mathcal{B}$ next update. The retained record then puts the older residual first within its movement. There is no waiting-target allocation or old-target rationing law.

h is the selected queue-perception estimate: pre-state local lane mass plus earlier actual admissions. It contains no invented waiting targets and does not subtract forecast signal discharge. Actual next occupancy separately includes that step's exits. Raw queue mass is the first perception model; no service-aware cost or unequal preference is added.

Access in the admitted layouts

For an unassigned pool with independently accessible receiving lanes, $E_{r\ell} = 1$ on those paths. Movement labels alone do not create access. A source feeding independent V0 paths has such access; a source or U feeding one physical F has only the edge into F and its shared limit.

For an occupied single lane, including F at its fork,

$$
E_{r\ell} = \mathbf{1}\{\text{every earlier resident cohort has fully crossed}\} \quad \text{on a legal outgoing edge.} \tag{9}
$$

An older physical residual prevents later physical entry even while the allocator inspects later records. Several actual lanes have separate ledgers/access conditions, not one global FIFO.

Movement-specific blocking is the logical scan rule; physical access after lane occupation is a separate constraint. At U–F, a full F restricts all movements because they share one receiving space. At the fork, a through head can obstruct a later turner despite an empty bay. Neither obstruction follows merely from the input record's order.

A zero left-turn stop-line rate is not automatically zero lane-entry room. A left turner can join an accessible shared queue and subsequently block through traffic. A full bay also does not forbid fresh left entry into a reachable shared continuation with room. Each constraint refers to its actual physical boundary.

Direct definition checks

The first case below is a local operator diagnostic, not a new primary scenario or a claim that every aggregate road offers independent access. For an independent-access pool with order $[L(0.4), T(0.6)]$, $A = 1$, $S = 0.6$ and exclusive-destination receiving $(0, 0.5)$, left entry is zero and through entry 0.5. Residual order is $[L(0.4), T(0.1)]$. Truncating candidates to the first S mass would expose only 0.2 through mass and give the wrong result for this pool rule.

For actual F take $[T(0.4), L(0.6)]$, $S_F = 1$, $R_T = 0.2$, $R_B = 1$. Through entry is 0.2, left entry zero. Reverse the order with the same totals and branch loads $(n_B, n_T) = (0, 1.8)$. Left uses B first, then through enters 0.2: total 0.8 rather than 0.2. These branch bounds are compatible with $\Delta t = \tau^f = \tau^b = 1$, $C_B = C_T = 1$, $K_B = K_T = 2$. They are prepared hand states, not a direct initializer or rollout.

A separate compatible balancing case has loads $(1.2, 1.0)$, receiving $(0.8, 1.0)$, ready left mass 1 and sending 0.6. Additions $(0.2, 0.4)$ leave perceived loads $(1.4, 1.4)$; the remaining upstream 0.4 is unassigned.

5 Elapsed service and selected receiving rates

A stop-line lane exposes its current ready resident head, with sending quota S_ℓ and cap $c_\ell = \mu_{\ell m}(g)$. The selected coarse receiving convention is

$$\rho_j = R_{j,k}/\Delta t. \tag{10}$$

This is a timestep computational average, not an inferred within-step physical supply curve.

For each output j, collect current heads by inbound leg i. A leg-to-output movement here has one service group:

$$C_{ij} = \sum_{\ell \in \mathcal{L}_{ij}} c_\ell, \quad w_{ij} = g_{p(i,j)}, \quad W_j = \min(\rho_j, \sum_i C_{ij}),$$

$$r_{ij} = \min(C_{ij}, \theta_j w_{ij}), \quad \sum_i r_{ij} = W_j, \tag{11}$$

$$r_\ell = \min(c_\ell, \eta_{ij}), \quad \sum_{\ell \in \mathcal{L}_{ij}} r_\ell = r_{ij}. \tag{12}$$

Remove zero-cap competitors; zero W gives zero rates. Each leg is weighted once; its ready lanes share equally with caps/reuse. Splitting a leg into more software lanes cannot multiply its leg priority.

Hold rates until the earliest head/quota exhaustion or step end, advancing all lanes together. For residual head mass q_ℓ,

$$\delta = \min\left\{\Delta t - \sigma, \ \min_{\ell: r_\ell > 0} \frac{\min(q_\ell, S_\ell^{\mathrm{rem}})}{r_\ell}\right\}. \tag{13}$$

Transfer $r_\ell \delta$, process tied events together, expose new heads and recompute sharing. Stop when rates vanish or time ends. A zero-rate head blocks its lane. Newly admitted mass is outside this resident service tranche.

A later ready head joins current competition; earlier discharge is not compensated to equalize final totals. Constant receiving removes a separate consumable-pool exhaustion rule, but head clearance and changing competitors remain sequential. Receiving reuse does not transfer one signal group's service to another.

For ready left mass q_L followed by sufficient through mass, with nonbinding receiving/physical caps,

$$y_L = q_L, \qquad y_T = s_T g_T \left(\Delta t - \frac{q_L}{s_L g_L}\right), \quad 0 < \frac{q_L}{s_L g_L} < \Delta t. \tag{14}$$

With unit timestep/saturations, $q_L = 0.2$ and $(g_L, g_T) = (0.4, 0.6)$, through discharge is 0.3. A small transfer from through to left gives through-discharge derivative 0.25 on this branch: faster head removal outweighs the smaller through rate. This is not a complete-TST benefit claim.

6 Paired transfers, histories and feasibility

A one-in/one-out internal edge transfers its ready prefix up to $\min(S_{\mathrm{up}}, R_{\mathrm{down}})$. Forks use allocation/access; the central junction uses elapsed service. An external source adds $D_{o,k}$ to its waiting record before admission, with budget $Q_{o,k} + D_{o,k}$. A sink removes at most sending and its declared allowance.

Each entry interface here has one sending owner. Shared receiving at the central signal node uses the simultaneous current-head sharing law, rather than an arbitrary serial order of inbound legs. All nodes read pre-state histories/ledgers. Each section's sending and receiving is spent once at its boundary. New admissions are appended after selecting resident service; today's exits create no immediate receiving credit. The cumulative update is

$$N^+_{e,k+1} = N^+_{e,k} + b_{e,k}, \quad N^-_{e,k+1} = N^-_{e,k} + y_{e,k}, \quad n_{e,k+1} = n_{e,k} + b_{e,k} - y_{e,k}. \tag{15}$$

Internal flow leaves one owner and enters the next with identical mass and route identity. Source admission removes source waiting; sink exit removes road mass. No residual is duplicated as lane stock.

For a receiving lane with total entrance $b > 0$, append children in their actual admission subsequence. A child at entrance rank $[r, r + q']$ gets

$$\left[t_k + \Delta t\, \frac{r}{b}, \ t_k + \Delta t\, \frac{r + q'}{b}\right], \qquad \text{eligibility shifted by } \tau_e^f. \tag{16}$$

For zero inflow create no segment. This is M1-style affine boundary reconstruction applied to the new selection rule, without service-event history knots. Serialize cross-input outgoing records by internal service-interval start, then fixed input-leg index, lane index and source cohort index. This orders macro bookkeeping, not rate grants or a new physical zipper rule. Those roads lead to sinks here; their scalar propagation/total time cannot acquire a later signal-order advantage from that serialization.

The timestep domain gives $S_{e,k} \le n_{e,k}$ and $R_{e,k} \le K_e - n_{e,k}$. Hence

$$0 \le b_{e,k} \le R_{e,k}, \quad 0 \le y_{e,k} \le S_{e,k} \implies 0 \le n_{e,k} + b_{e,k} - y_{e,k} \le K_e. \tag{17}$$

Paired transfers and route-preserving splits imply

$$\sum_o Q_{o,k} + \sum_e n_{e,k} + Z_k = D_{<k}, \tag{18}$$

also route by route. Z_k is cumulative sink exit; $D_{<k}$ all earlier exogenous release.

A finite state has a defined unique result: finite source scan, unique convex allocation vectors, unique capped-sharing vectors and a finite head-event calculation. Each positive service interval exhausts a head/quota or reaches step end. Exact ties are handled together; no mass cutoff or software priority manufactures uniqueness. These are definition-level arguments, not implementation or traffic-fidelity evidence.

7 Sensitivities of the revised update

Use a fixed structural order and active physical/constraint signature. Differentiate continuous masses, admissions, queues, histories and head times. Labels, permissions, access branches and the source-order skeleton are structural; their switches require branch/one-sided treatment.

For free allocation set $\mathcal{A}$ and upper-capped set $\mathcal{U}$,

$$\lambda = \frac{V - \sum_{\mathcal{U}} u_\ell + \sum_{\mathcal{A}} h_\ell}{|\mathcal{A}|}, \qquad \dot\lambda = \frac{\dot V - \sum_{\mathcal{U}} \dot u_\ell + \sum_{\mathcal{A}} \dot h_\ell}{|\mathcal{A}|},$$
$$\dot x_\ell = \begin{cases} \dot\lambda - \dot h_\ell, & \ell \in \mathcal{A}, \\ \dot u_\ell, & \ell \in \mathcal{U}, \\ 0, & \text{inactive lower branch.} \end{cases} \tag{19}$$

On stable fully capped branches $\dot x = \dot u$; stable zero flow has $\dot x = 0$. Differentiate V through its active minimum and u through its actual residual receiving/access branch. The scan propagates

$$\dot h \leftarrow \dot h + \dot x, \quad \dot R^{\mathrm{rem}} \leftarrow \dot R^{\mathrm{rem}} - \dot x, \quad \dot s \leftarrow \dot s - \dot V, \quad \dot q^{\mathrm{rem}} = \dot q - \dot V. \tag{20}$$

There is no frozen-waiting-vector derivative. A blocked cohort's zero admission does not erase its retained mass sensitivity. Movement and physical-prefix switches are not smoothed.

For uncapped leg set $\mathcal{F}$ and capped set $\mathcal{C}$,

$$\dot\theta = \frac{\dot W - \sum_{\mathcal{C}} \dot C_{ij} - \theta \sum_{\mathcal{F}} \dot w_{ij}}{\sum_{\mathcal{F}} w_{ij}}, \qquad \dot r_{ij} = \begin{cases} \dot C_{ij}, & i \in \mathcal{C}, \\ w_{ij}\dot\theta + \theta \dot w_{ij}, & i \in \mathcal{F}. \end{cases} \tag{21}$$

If the uncapped set is empty, differentiate the cap branches directly; a stable zero-supply branch has zero grants. No empty-set division is used. Within-leg sharing uses the unweighted analogue, with the same boundary cases. For an interior event $\delta = q/r$, $\dot\delta = \dot q/r - q\dot r/r^2$; the final interval derivative is minus the preceding duration derivatives. Thus

$$y = \sum_a r_a \delta_a, \qquad \dot y = \sum_a (\dot r_a \delta_a + r_a \dot\delta_a). \tag{22}$$

Head times cannot be frozen merely because outputs are evaluated at timestep boundaries.

Fixed delayed queries have fixed interpolation weights but sensitive history values. For an interior eligibility cut at fixed query t,

$$f = \frac{t-a}{b-a}, \quad \dot f = \frac{-\dot a - f(\dot b - \dot a)}{b-a}, \quad \dot q^{\mathrm{ready}} = f\dot q + q\dot f. \tag{23}$$

Use whole/empty branches outside $[a, b]$; point segments need their boundary branch. For positive entrance b,

$$\dot{t}_{\text{front}} = \Delta t \left(\frac{\dot{r}}{b} - \frac{r\dot{b}}{b^2} \right), \tag{24}$$

with the analogous tail response. No epsilon repairs zero inflow. Exact-compatible merging must preserve the physical map; a storage-format change is not automatically a physical kink.

8 Complete population time and continued state response

Collect physical stocks, histories, ledgers and sources in X_k, with revised update $X_{k+1} = F_k(X_k, u_k)$. The inherited cost and its response are

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} \left[\sum_o (Q_{o,k} + D_{o,k}) + \sum_e n_{e,k} \right], \tag{25}$$

$$P_0 = 0, \quad P_{k+1} = F_{k,X} P_k + F_{k,u} \dot{u}_k, \quad \dot{J} = \sum_k (c_{k,X} P_k + c_{k,u} \dot{u}_k). \tag{26}$$

Within control $\dot{u}_k = d$; in fixed common continuation $\dot{u}_k = 0$, while P_k keeps evolving. Empty loading/common preparation is independent of the perturbed candidate; otherwise propagate its response too. No per-vehicle trajectory gradient is constructed.

At fixed state, the immediate cost partial for current service is zero. Fork admission also has no direct green input at fixed pre-state: its feedback comes through earlier queues/history. Stop-line service can respond directly now. These effects compose into access, spillback, source waiting and eventual sink exit. Internal transfer alone does not remove population. For fixed releases,

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} (D_{\le k} - Z_k), \qquad \dot{J}_{\Delta t} = -\Delta t \sum_{k=0}^{H-1} \dot{Z}_k. \tag{27}$$

Use a common integer horizon qualified to clear the compared action neighborhood, followed by zero occupancy. Finite demand, acyclic routes, positive continuation service and eventual sink availability support qualification; no universal numerical horizon or clearance theorem is asserted. A changing stop counter, residual cutoff or nonclearing case cannot silently supply complete TST. The derivative domain includes clearance and unchanged active signatures; switches need directional branches, not a global smoothness claim.

9 Matched comparisons and native qualification

Models share population, geometry/permissions, action, objective, preparation, continuation and a declared data protocol. Original fine-grained series may pass through each model's own macro converter/estimator. Declare retained variables, memory, update schedule and calibration data. Native subsets are legitimate. Successful macro inference counts against needing exact order; an exact ordered ledger retained inside an estimator remains meso information.

Equal current totals do not establish equality of all lawful histories. State-conditioned replay, observation-updated prediction and forecasting from a common prefix are distinct tasks. State which observations are fixed or regenerated for each derivative claim. A later observation cannot count as a response predicted before it was observed.

Baseline roles remain B0 aggregate, B1 structured movement queues, B2 responsive lane/storage aggregates and B3 this selective model. They are not a universally nested information ladder. Qualify principal instances by soundness/adaptation, task fit and then ease.

B1's working source is Wei et al.'s link queue model, with moving/queued state and native effective green. Its full causal short-region/link/node law must be instantiated; the fixed-speed shortcut is not assumed valid in every spillback regime.

B2's working direction uses Gong et al.'s lane-exit allocation with a sourced storage adaptation, for which Liu–Chang (2011) is a candidate. Exit allocation is not finite-bay entry, and native theorems do not

automatically qualify that composition. Exact implementations/adapters remain pending under feedback 8; this report does not declare them qualified.

The old exclusive-target homogeneous-mixture ablation is unselected and archived, not carried into the chosen shared-permission case. No weaker substitute is adopted to finish the report. A V2 comparator failure must be understood before attributing a V3 advantage to retained order.

10 Definition checks and external usefulness

R1 covers pool/F access, partial admission, finite storage, current-head sharing/reuse, route conservation and the source-counted objective. V0 neutral cases need matched pure service classes and nonbinding or equivalent receiving; one through/right green does not make differently constrained exits identical. In that neutral domain the laws reduce to the same sending/service/receiving minimum. Exact equivalence concerns a matched aggregate backbone; a source-native propagation difference must be qualified and cannot be credited to retained order.

Rotating the symmetric scenario and green groups must rotate results. Equal lane totals do not justify one larger FIFO lane without corresponding order, readiness, access, service and receiving. Hand/reference checks establish definition correctness, not realism.

For R2 qualify one fixed executable mapping. With fixed cycle C^{cyc}, lost intervals totaling $L < C^{\mathrm{cyc}}$, order and offset,

$$G_p = (C^{\mathrm{cyc}} - L)g_p, \qquad s^{\mathrm{eff}}_{\ell m} = s^{\mathrm{phys}}_{\ell m}(C^{\mathrm{cyc}} - L)/C^{\mathrm{cyc}}. \tag{28}$$

Declare program admissibility and the common switch to fixed clearance; do not silently insert minimum green to repair an API. This matches nominal average opportunity, not microscopic startup/access or every phase effect. Official SUMO signal definitions support the interface, not empirical equivalence.

Use a finite master route population and common desired releases. Integer reference vehicles are exogenous population, not rounded internal mass. Fix native lane behavior independently of candidate directions; do not replay B3's private assignments into only one model. Use matched predeclared reference seeds and held-out family IDs, keeping order pairs together.

Reference time is desired-release-to-arrival time: `duration` plus `departDelay`, from official TripInfo. Waiting is already inside duration. Retain uninserted/unfinished members in master-population accounting; clear all demand or report noncompletion, never a completed-trip-only objective.

For each feasible direction, compare predicted sign/ranking and scale with actual richer-evaluator complete-time change under the same action mapping. Disclose observation conditioning. Only after forward/reference and derivative checks pass should matched optimization and held-out usefulness be tested.

11 What the consolidation establishes

The closed policies now define eligible source-order scanning, movement-specific pending priority, actual-admission balancing, physical lane access, elapsed service, fixed histories and complete cost sensitivity. The scan/FIFO conflict is resolved by actual owner/access, not by deleting the feeder or inventing overtaking.

This supplies feasibility/uniqueness reasoning, analytic branches and hand checks. Implementation correctness, AD/finite-difference agreement, qualified native rivals and external usefulness remain future evidence. No simulator, numerical-gradient, optimization or SUMO experiment is run.

The source map credits every law as inherited, adapted, defined here or derived. M0 attributes its backbone to Yperman (2007) and UNsim v0.12.0; this pass checks the approved repository transcription, not fresh full reproduction of those originals. M1 supplies route/timing/merge machinery; the non-prefix unassigned-pool scan is an explicit urban adaptation. No publication is credited with the whole model.